Duration: **90 minutes**
Aids Allowed: **NONE**

Student Number: |_|_|_|_|_|_|_|_|_|_|

Last (Family) Name: ______________________________

First (Given) Name(s): ______________________________

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*Do **not** turn this page until you have received the signal to start.*
(In the meantime, please fill out the identification section above, and read the instructions below *carefully*.)

This term test consists of 5 questions on 6 pages (including this one), *When you receive the signal to start, please make sure that your copy of the test is complete.*

Answer each question directly on the test paper, in the space provided, and use the reverse side of the last page for rough work. If you need more space for one of your solutions, use the reverse side of the final page and *indicate clearly the part of your work that should be marked.*

General Hints: We were careful to leave ample space on the test paper to answer each question. Note that the exam is worth 50 marks, so if you average more than 2min per mark, you may run out of time.

MARKING GUIDE

1: ______/15

2: ______/ 5

3: ______/10

4: ______/10

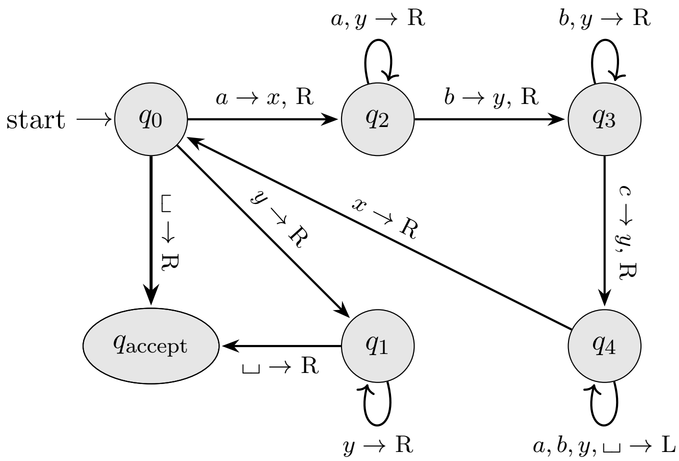
5: ______/10

TOTAL: ______/50

Good Luck!

Question 1. [15 MARKS]

Consider the following Turing machine:



If this machine is run on the input $aabbbc$, its first configuration will be $q_0aabbbc$. What will the remaining configurations be? Fill in the chart below.

C_0: $\mathbf{q_0\,a\,a\,b\,b\,b\,c\,\sqcup}$	C_5: $x\,a\,y\,b\,b\,q_3\,c\,\sqcup$	C_{10}: $x\,a\,q_4\,y\,b\,b\,y\,\sqcup$	C_{15}: $x\,x\,y\,q_2\,b\,b\,y\,\sqcup$
C_1: $x\,q_2\,a\,b\,b\,b\,c\,\sqcup$	C_6: $x\,a\,y\,b\,b\,y\,q_4\,\sqcup$	C_{11}: $x\,q_4\,a\,y\,b\,b\,y\,\sqcup$	C_{16}: $\mathbf{x\,x\,y\,y\,q_3\,b\,y\,\sqcup}$
C_2: $x\,a\,q_2\,b\,b\,b\,c\,\sqcup$	C_7: $x\,a\,y\,b\,b\,q_4\,y\,\sqcup$	C_{12}: $\mathbf{q_4\,x\,a\,y\,b\,b\,y\,\sqcup}$	C_{17}: $x\,x\,y\,y\,b\,q_3\,y\,\sqcup$
C_3: $x\,a\,y\,q_3\,b\,b\,c\,\sqcup$	C_8: $x\,a\,y\,b\,q_4\,b\,y\,\sqcup$	C_{13}: $x\,q_0\,a\,y\,b\,b\,y\,\sqcup$	C_{18}: $x\,x\,y\,y\,b\,y\,q_3\,\sqcup$
C_4: $\mathbf{x\,a\,y\,b\,q_3\,b\,c\,\sqcup}$	C_9: $\mathbf{x\,a\,y\,q_4\,b\,b\,y\,\sqcup}$	C_{14}: $x\,x\,q_2\,y\,b\,b\,y\,\sqcup$	C_{19}: $x\,x\,y\,y\,b\,y\,\sqcup\,q_R\,\sqcup$

Question 2. [5 marks]

Consider the language

$$L_5 = \{\langle M, x\rangle \mid \text{For any TM } N\text{, if } M \text{ accepts } \langle N\rangle\text{, then } N \text{ loops on } x\}.$$

Write L_2 in set notation using only quantifiers and decidable predicates.

Solution:

$$L_2 = \{\langle M, x\rangle \mid \forall\langle N, k\rangle, \langle M, N, k\rangle \in ACC \to \langle N, x, k\rangle \notin H\}.$$

If the quantifiers are correct and the predicates are correct but written in English, then I'll accept the answer. Stating that N is a TM and that k is in $\mathbb{N}$ is also OK.

Question 3. [10 marks]

Let L_3 be any co-recognizable language. Prove that $L_3 \leqslant_m E_{TM}$.

Solution:

Let L_3 be any co-recognizable language. Then $\overline{L_3}$ is recognizable, and so it has a recognizer R_3.

Here's our reduction:

Let $P =$"On input $\langle x\rangle$:

1. Let $M' =$"On input $\langle w\rangle$:
    1. Run R_3 on $\langle x\rangle$.
    2. If it rejects, *loop*.
    3. Otherwise, *accept*."
2. Return $\langle M'\rangle$."

Proof that $\langle x\rangle \in L_3$ iff $\langle M'\rangle \in E_{TM}$:

$\Rightarrow$) **Suppose that** $\langle x\rangle \in L_3$:

Then $x \notin \overline{L_3}$.

So R_3 will not accept $\langle x\rangle$ when it is run on step 1 of M'.

Therefore, either R_3 loops on step 1 of M' or it rejects, in which case M' loops on step 2.

Either way, M' does not accept.

Since this is true for any input $\langle w\rangle$ for M', $L(M') = \varnothing$.

$\Rightarrow \langle M'\rangle \in E_{TM}$.

$\Leftarrow$) **Suppose that** $\langle x\rangle \notin L_3$**:**

Then $x \in \overline{L_3}$.

So R_3 will accept $\langle x\rangle$ when it is run on step 1 of M'.

So M' will accept on step 3.

Since this will occur for every input w for M', $L(M') = \Sigma^* \neq \varnothing$.

$\Rightarrow \langle M'\rangle \notin E_{TM}$.

Question 4. [10 MARKS]

Consider the language L_4:

$$L_4 = \big\{ \langle M \rangle \ \big| \ \ M \text{ does not reject more than two strings } \big\}.$$

Prove that L_4 is not recogizable.

Solution:

We'll reduce from $\overline{HALT}$.

Consider the TM P "On input $\langle M, w \rangle$:

1. Let $M' =$ "On input $\langle x \rangle$:
   1. Run M on $\langle w \rangle$.
   2. *Reject*."
2. Return $\langle M' \rangle$."

Proof that $\langle M' \rangle \in L_4$ iff $\langle M, w \rangle \in HALT$:.

($\Leftarrow$) Suppose that $\langle M, w \rangle \in \overline{HALT}$.
Then M loops on $\langle w \rangle$.
So for any input $\langle x \rangle$, M' loops on step 1.
This means that M' never reaches step 2, and so never rejects an input.
$\Rightarrow \langle M' \rangle \in L_4$.
($\Rightarrow$) Suppose that $\langle M, w \rangle \notin \overline{HALT}$.
Then M halts on $\langle w \rangle$.
So for any input $\langle x \rangle$, M' eventually finishes step 1.
This means that M' always reaches step 2, and so always rejects its input.
So the number of strings M' rejects is $|\Sigma^*| = \infty > 2$.
$\Rightarrow \langle M' \rangle \notin L_4$.

Question 5. [10 MARKS]

Consider the language L_4 from the previous question:

$$L_4 = \{\langle M \rangle \mid \ M \text{ does not reject more than two strings } \}.$$

Is L_4 co-recognizable? Prove your answer.

Solution:

L_4 is co-recognizable. To see why, we can write $\overline{L_4}$ as:

$$\begin{aligned}\overline{L_4} &= \{\langle M \rangle \mid \ M \text{ rejects more than two strings } \} \\ &= \{\langle M \rangle \mid \ M \text{ rejects at least three strings } \} \\ &= \{\langle M \rangle \mid \exists \langle x, y, z, k\rangle, x \neq y, x \neq z, y \neq z \text{ and } M \text{ rejects } x,\ y, \text{ and } z \text{ in } k \text{ steps } \}.\end{aligned}$$

So we just need to find three strings that M rejects.

With that in mind, here's our recognizer for $\overline{L_4}$:

Let $x_0, x_1, \ldots$ be an effective enumeration of Σ^*.

Let $R_4 =$"On input $\langle M \rangle$:
1. Fo $k = 0$ to ∞:
2. For $i = 0$ to k:
3. For $j = 0$ to k:
4. For $\ell = 0$ to k:
5. If $x_i \neq x_j, x_i \neq x_\ell$, and $x_j \neq x_\ell$:
6. Run M on each of x_i, x_j, and x_ℓ for k steps.
7. If all three reject, *accept.*

Proof that R_4 accepts $\langle M \rangle$ iff $\langle M \rangle \in \overline{L_4}$:

Suppose $\langle M \rangle \in \overline{L_4}$:

Then there are at least three strings that M rejects.

So we can choose one of these triples, say, $(x_{i'}, x_{j'}, x_{\ell'})$, where $x_{i'}, x_{j'}$, and $x_{\ell'}$ are all different strings.

We let k' be the maximum number of steps M takes to reject all three strings.

Then, if we run R_4 on $\langle M \rangle$ and it reaches the loop iteration $k = \max(k', i', j', \ell^{prm}), i = i', j = j', \ell = \ell'$, we see that M will reject all three strings on line 6.

In this case, M will accept on line 7.

Note that R_4 can't loop inside of any loop iteration, since there are time limits when running M. So if we don't reach this loop iteration, it means that M has already accepted.

$\Rightarrow R_4$ accepts $\langle M \rangle$.

Suppose $\langle M \rangle \notin \overline{L_4}$:

Then, for any triple $(x_{i'}, x_{j'}, x_{\ell'})$, where $x_{i'}, x_{j'}$, and $x_{\ell'}$ are all different strings, M will not reject all three of $x_{i'}$, $x_{j'}$, and $x_{\ell'}$, regardless of how long we run it.

Suppose we run R_4 on $\langle M \rangle$. We can see that for any loop iteration k, i, j, ℓ, either (x_i, x_j, x_ℓ) are not distinct strings (in which case we don't run lines 6 and 7), or M fails to reject at least one of these strings in k steps (in line 6).

Either way, M will not accept on line 7.

Since this is true for all loop iterations, R_4 will never accept.

So R_4 does not accept $\langle M \rangle$.